Three Generations of Healthcare IT: From the Digital Record to the Computable Care Process

Alexander Apartsin[1], Yehudit Aperstein[2]

[1]School of Computer Science, Faculty of Sciences, Holon Institute of Technology (HIT), 52 Golomb St., Holon 5810201, Israel

[2]Intelligent Systems, Afeka Academic College of Engineering, 218 Bnei Efraim St., Tel-Aviv 6910717, Israel

Correspondence: Yehudit Aperstein, apersteiny@afeka.ac.il. Co-author: A. Apartsin, alexanderap@hit.ac.il. ORCID: Apartsin 0009-0000-7007-3529, Aperstein 0000-0001-6390-9463. *Submitted to JAMIA as a Perspective.*

Abstract

Objective. Healthcare information technology is usually organized by the technologies it adopts. We instead organize it by the unit of information a system makes computable, and describe an emerging computational layer whose object is patient-specific clinical intent.

Approach. We give criteria for a computational layer, derive three (record, clinical state, and a proposed layer of clinical intent), and formalize the Actionable Clinical Record (ACR) as the atomic object of the third layer.

Discussion. The framework distinguishes prescribed, observed, and intended process; existing standards represent intent once it is structured but do not recover it from natural communication, the capability we localize. The ACR is complementary to FHIR workflow resources, guidelines, and process mining; a companion feasibility study supports tractability.

Conclusion. Computable clinical intent is a coherent research direction; the ACR, its maturity ladder, and an executable-correctness evaluation framework are reusable constructs that subsequent work can extend, evaluate, or falsify.

Keywords: electronic health records; clinical workflow; actionable clinical record; clinical intent; natural language processing; large language models; FHIR; care-process automation

1. Introduction

Healthcare information technology can be analyzed more durably through the computational objects it supports than through the technologies it adopts, which change rapidly.[1,2] Organized this way, the field's history is compact: the first generation made the clinical *record* computable, the second the clinical *fact*, and a third, which we propose, makes patient-specific clinical *intent* computable from communication. We use *generation* for the historical emergence of a new computational layer; the layers accumulate rather than replace one another.

A limitation persists after the first two layers are in place. When a clinician writes "repeat the complete blood count in two weeks" or "refer to cardiology if symptoms persist," the instruction is decisive yet captured only as narrative text; its action, timing, condition, and responsible actor remain outside the computable substrate. The cost is measurable: test-result follow-up in ambulatory care often fails,[3] only about one-third of specialist referrals reach a documented completed visit,[4] and such loop-closure failures contribute to diagnostic error and avoidable harm.[5,6]

This Perspective offers a framework and a vocabulary rather than a new empirical study: criteria for a computational layer and a three-layer organization of the field (Section 2); a formal definition of the Actionable Clinical Record (ACR) as the atomic object of the third layer (Section 5); an executable-correctness evaluation framework with a companion feasibility study (Section 7); and a research agenda (Section 8).

2. A Framework of Computational Layers

We distinguish a computational layer from an incremental trend by five markers: a new atomic *unit of computability*; an *external driver*; an *enabling technology*; a new *class of computation* the unit permits; and a *residual limitation* that motivates the next layer's object; a candidate that fails these markers is an incremental capability within an existing layer, not a new one. The layers are *cumulative*, each building on those beneath it (Figure 1), and differ in epistemic status: the first two are *retrospective abstractions* over a well-documented history of electronic health records,[1,2] whereas the third is a *prospective hypothesis* offered as an analytic lens rather than an established periodization. The lens is complementary to the data–information–knowledge hierarchy[7] and to digital-maturity models such as EMRAM,[8,9] which measure how much capability an institution has adopted; our concern is *which* clinical object becomes computable (Table 1).

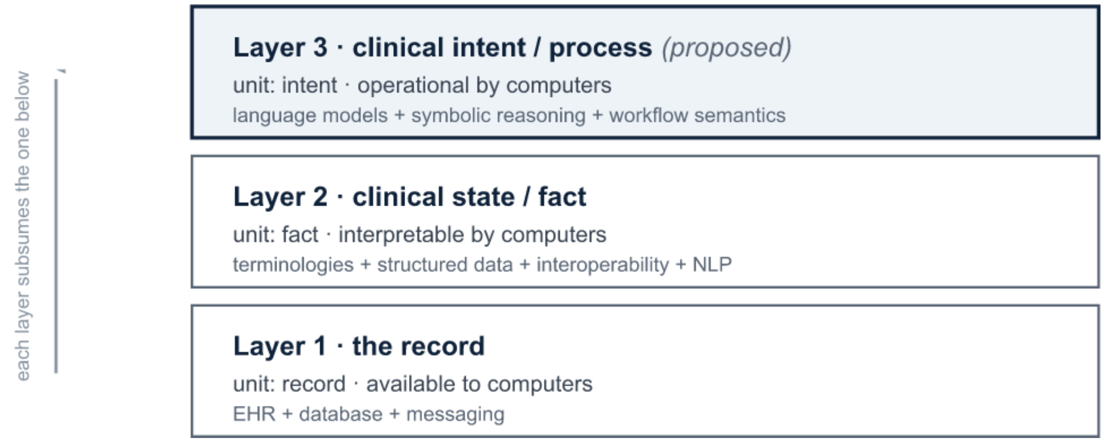

Figure 1. *The three computational layers of healthcare IT, organized by the unit of information each makes computable. Layers accumulate rather than replace; layer 3 is proposed. Alt text: three stacked boxes, record at the base, clinical state above it, and clinical intent at the top, with an upward arrow indicating that each layer subsumes the one below.*

Axis	Layer 1: Record	Layer 2: Clinical state	Layer 3: Clinical intent (proposed)
Unit of computability	Clinical record	Discrete clinical fact	Intended clinical action (ACR)
Question answered	What was documented?	What is true about the patient?	What is supposed to happen?
External driver	Adoption policy and reimbursement	Interoperability and quality measurement	Care continuity and patient safety
Enabling technology	Transactional databases; messaging	Terminologies; structured data; interoperability; clinical NLP	Language models; symbolic reasoning; workflow semantics
Class of computation	Store, retrieve, exchange	Query, reason, predict	Schedule, coordinate, monitor, close
Canonical output	Document	Coded fact	Actionable Clinical Record
Residual limitation	Meaning remains in narrative	Intent remains in communication	Safe, reliable execution

Table 1. *The three computational layers against the framework markers. Layers 1 and 2 are retrospective; layer 3 is proposed. The residual limitation of each layer motivates the unit of the next.*

3. The First Two Layers: Record and Clinical State

The first layer digitized the clinical record and its transactions, replacing the paper chart with electronic capture, storage, retrieval, transmission, and operations such as order entry and results routing. In the United States, HITECH and Meaningful Use moved adoption to near-universal within a decade,[10,11] extending the older problem-orientedrecord ideal of an organized, auditable chart.[12] Its enabling technology, the transactional database, stored and moved information without making its clinical meaning computable across systems.

The second layer turned the record into computable clinical *state*. Its enabling stack was broad and cannot be reduced to a single technique: structured data entry, controlled terminologies,[13] computerized order entry and coded lists, clinical data warehouses, and interoperability standards[14,15] were as consequential as language processing, which was one route by which narrative facts entered the semantic layer (MedLEE and cTAKES parsing reports into coded concepts).[16,17] These enabled the first broad wave of computation *on* clinical data: decision support, quality measurement, registries, and research.[18] A tension remained: encoding meaning into structured fields captures facts, while much of the reasoning, and nearly all of the intended action, stays in narrative.[19]

4. The Remaining Computational Gap

The first two layers made clinical *state* computable but not the *intended actions* that constitute care. Healthcare IT can represent process, but only once a person or system has structured it: FHIR represents patient-specific intent through *CarePlan*, *ServiceRequest*, and *PlanDefinition*;[14] computer-interpretable guidelines encode executable protocols;[20,21] and process mining reconstructs pathways from event logs.[22,23] These paradigms are complementary to the proposed layer and often provide the infrastructure to operationalize its outputs. The gap is isolated by distinguishing three kinds of process information (Table 2): *prescribed* (what should generally happen), the domain of guidelines and decision support; *observed* (what actually happened), the domain of records and process mining; and *intended* (what is supposed to happen for a specific patient), expressed primarily in communication. Existing paradigms address the first two. The missing capability, and the subject of the third layer, is recovering intended process from ordinary communication and converting it into computable, executable form.

Process type	Meaning	Typical source	Existing paradigm
Prescribed	What should generally happen	guideline / protocol	computer-interpretable guidelines; decision support
Observed	What actually happened	event log / EHR	process mining
Intended	What is supposed to happen for a specific patient	clinical communication	proposed third layer

Table 2. *Three kinds of process information. The proposed third layer targets intended process, which the prescribed and observed paradigms do not recover from natural communication.*

5. Computable Clinical Intent

We propose that the third layer makes patient-specific clinical intent computable by recovering it from communication as a structured, executable representation. We use one terminology throughout: clinical *communication* expresses clinical *intent*; an intent decomposes into intended *actions*; each action is an *Actionable Clinical Record*; a set of ACRs and related events is a clinical *workflow*; and the enacted sequence over time is the *care process*.

5.1 The Actionable Clinical Record

Definition 1 (Actionable Clinical Record). An Actionable Clinical Record is a source-grounded representation of a patient-specific intended clinical action, with the semantic attributes required to interpret, execute, monitor, or audit it: the tuple ACR = ⟨ action, target, actor, temporal constraint, condition, dependency, status, provenance, confidence ⟩

in which *action*, *status*, and *provenance* are required, and the remaining attributes are populated when the communication supports them.

The attributes are defined semantically (Figure 2). Beyond *action*, *target* (linked where possible to a layer-2 coded concept), and the responsible *actor*, the *temporal constraint* is a normalized time semantics rather than a date alone, admitting absolute ("within six months"), relative ("before the next visit"), event-anchored ("after finishing antibiotics"), and recurring forms; *condition* is a clinical trigger; *dependency* encodes ordering; *status* is the rung on the maturity ladder; *provenance* grounds the record in its source span, speaker, and encounter; and *confidence* is a calibrated per-attribute uncertainty routing uncertain records to review.

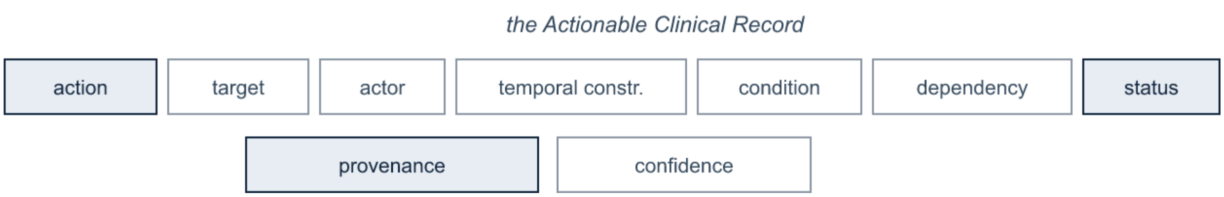

Figure 2. *The ACR schema. Alt text: nine labeled boxes forming the ACR tuple; action, status, and provenance are shaded to mark them as required; target, actor, temporal constraint, condition, dependency, and confidence are outlined as optional.*

The third layer is a capability ladder, and naming its rungs prevents over- and under-claiming. Recognizing that "complete blood count" appears in a note belongs to layer 2; extracting "repeat complete blood count" is a partial step; linking it to "two weeks" and a responsible actor produces an ACR; mapping it to a computed due date and a workflow object makes it executable; and detecting completion or escalation closes the loop. We label the rungs *mentioned, interpreted, actionable, executable, monitored,* and *closed*; the ladder doubles as the ACR's *status* attribute and distinguishes an *actionable* record from an *executable* one carrying enough resolved information to act on automatically.

5.2 From communication to workflow

The third layer is a bridge rather than a replacement electronic health record (Figure 3). Its inputs are the range of clinical communication (discharge instructions, conversation, handovers, referrals, portal messages, telephone calls, ambient transcripts); its output is a set of ACRs that maps onto infrastructure the field already has.[14] The mapping is explicit and situates the ACR *upstream* of these resources rather than as a competitor: *action* and *target* populate a *Task* or *ServiceRequest* and its code, *temporal constraint* the occurrence timing, *condition* a trigger or precondition, *actor* the owner or performer, *dependency* the *basedOn*/*partOf* links, and *status* the resource lifecycle, while *provenance* and *confidence* are extraction metadata with no native slot. Existing systems execute such instructions once structured; the capability the third layer adds is recovering them when they originate as natural communication.

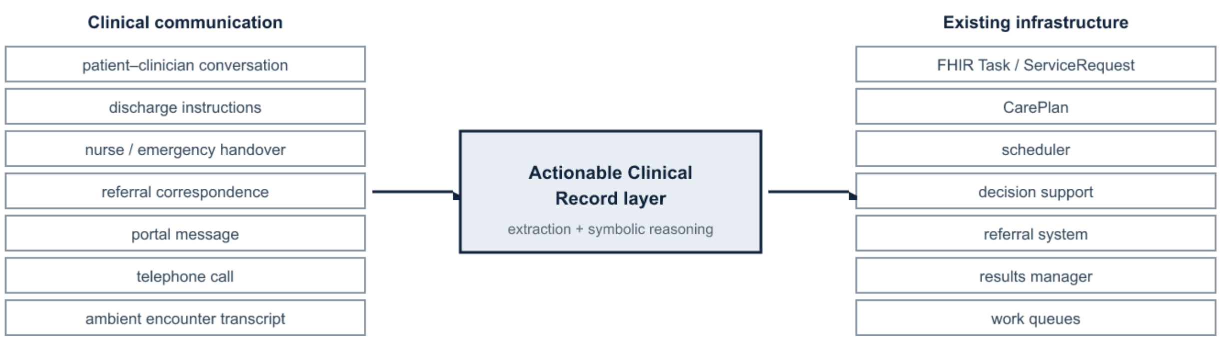

Figure 3. *The ACR as the layer between clinical communication and existing workflow infrastructure. Alt text: communication sources on the left feed a central Actionable Clinical Record layer, which maps to existing infrastructure such as FHIR resources, schedulers, and referral and results systems on the right.*

Follow-up is the most-documented case, but it is one of a broader class of communication-derived process objects sharing the ACR representation: conditional escalation (“return immediately if fever develops”), medication transitions (“reduce the dose after seven days”), handoff commitments, referrals, pending-result responsibility, and patient self-management. Recent work on patient messages, incidental-finding follow-up, and ambient documentation indicates these channels routinely carry clinically consequential actionable content.[24]

5.3 Relation to existing paradigms

Two mature lines can be mistaken for the third layer. Clinical NLP already extracts concepts, relations, events, and temporality;[16,17,25] the distinction is not facts versus actions but operational workflow semantics and execution reliability, the requirement that a recovered action be source-grounded, temporally normalized, and reliable enough to act on rather than only index. Advances in language models improve the feasibility of recovering these relations, but improving extraction is not, by itself, a layer transition. Computer-interpretable guidelines encode generic, population-level protocols;[20,21] the third layer recovers patient-specific records from a particular encounter. The two are complementary: guidelines state what should generally happen; the third layer captures what a clinician instructed for this patient.

6. Reliability and Architecture

A more capable text generator does not remove the requirements the third layer imposes. Auditability, verifiable temporal correctness, and a calibrated signal of when to defer are properties of the output contract, not gaps in model capability: a system acting on patient trajectories must establish that a computed date is correct and trace it to its source. General-purpose models interpret clinical language well but are less reliable where the layer is most demanding, in binding an action to its time and normalizing it into an executable schedule without omission or fabrication.[26,27]

These requirements motivate a hybrid decomposition in which learned components interpret and explicit components enforce verifiable structure.[28] Because some workflow semantics (calendar arithmetic, dependency ordering, status transitions) are formally specified, learned extraction supplies candidate actions and arguments[29] while deterministic reasoning resolves times and due dates[30,31,32] and selective prediction routes uncertain cases to a clinician.[33] The requirements can be stated as invariants: every acted-upon record traces to a source span, speaker, and encounter; deterministic semantics are computed, not generated; each inferred attribute carries a calibrated confidence; the system may abstain; higher-consequence actions receive tighter human review;[33] an immutable trace links communication to action for audit;[34] and extracted content stays distinct from inferred. Reliability, not scale, is the governing constraint. Human confirmation is the expected operating mode until task-specific calibration and prospective safety evidence justify higher automation.

7. Evaluation Framework and Feasibility

Because the third layer acts, it should be evaluated on *executable correctness* rather than text overlap. We propose an evaluation framework for ACR recovery whose measures are reported separately: action detection; argument extraction; action–time, condition, and actor linking; temporal-normalization error; whole-record exact match; unsupported- and omitted-action rates; calibration and selective-risk behavior; source-provenance accuracy; and loop-closure outcome. This framework, rather than any single score, is the contribution we intend others to adopt.

A companion study offers a formative feasibility test of one minimal task, follow-up-instruction extraction, evaluating a hybrid neural-symbolic pipeline on a controlled benchmark against generative baselines.[35] On the action–time pairing the executable representation requires, the hybrid pipeline substantially outperforms general-purpose language models, which recognize actions but bind them to times unreliably. This is an existence proof for one narrow subproblem, not evidence for the framework; the framework's value rests on the constructs it defines and their evaluation across tasks.

8. Research Agenda

The framework defines a program in four domains. **Representation:** a consensus ACR ontology, with attribute semantics, constraint types, provenance, and reconciliation when communications revise or revoke a plan. **Inference and reliability:** ACR recovery from multilingual and multimodal communication, temporal reasoning, cross-message reconciliation, calibrated abstention, and distinguishing patient from clinician intent. **Evaluation:** authentic-data benchmarks beyond synthetic corpora, the executable-correctness framework of Section 7 adopted across tasks, and cross-institution and cross-language generalization. **Integration and impact:** ACR-to-FHIR mapping and write-back into scheduling, results-management, and referral systems,[14,20] human-factors design of the review queue, regulatory classification, and whether closed-loop operation improves clinical outcomes.

Moving from synthetic corpora to de-identified real-world notes across languages and documentation styles is the central empirical risk, and safe deployment presupposes the invariants of Section 6. The proposal is falsifiable: machine-actionable use of FHIR workflow resources should grow relative to free-text follow-up; evaluation should shift toward executable-correctness measures; reliable systems should expose source-linked, auditable records with selective review rather than end-to-end generation; and ambient documentation should extend from notes to tasks and orders.[24,36] Failure of these trends would count against it.

9. Conclusion

Healthcare IT has made the record and then the clinical fact computable; we propose patient-specific clinical intent as the next layer. What characterizes it is not generating clinical text but recovering, from ordinary communication, source-grounded and verifiable representations of intended action, formalized as the Actionable Clinical Record and situated upstream of existing workflow infrastructure. We offer the framework, the ACR, its maturity ladder, and the evaluation framework as reusable constructs that subsequent work can operationalize, evaluate, extend, or falsify.

Data Availability

No new data were generated for this article. The feasibility results discussed in Section 7 derive from a companion study;[35] details of that benchmark are reported therein.

Author Contributions (CRediT)

A. Apartsin: Conceptualization, Methodology, Writing – original draft. Y. Aperstein: Conceptualization, Supervision, Writing – review and editing.

Funding

None declared.

Conflicts of Interest

None declared.

References

1. Ammenwerth E, et al. Twenty-Five Years of Evolution and Hurdles in Electronic Health Records and Interoperability in Medical Research: Comprehensive Review. *J Med Internet Res* 2025;27:e59024. doi:10.2196/59024.

2. Evans RS. Electronic Health Records: Then, Now, and in the Future. *Yearb Med Inform* 2016;25(Suppl 1):S48–S61. doi:10.15265/IYS-2016-s006.

3. Callen JL, Westbrook JI, Georgiou A, Li J. Failure to follow-up test results for ambulatory patients: a systematic review. *J Gen Intern Med* 2012;27(10):1334–1348. doi:10.1007/s11606-011-1949-5.

4. Patel MP, Schettini P, O'Leary CP, Bosworth HB, Anderson JB, Shah KP. Closing the Referral Loop: an Analysis of Primary Care Referrals to Specialists in a Large Health System. *J Gen Intern Med* 2018;33(5):715–721. doi:10.1007/s11606-018-4392-z.

5. Singh H, Meyer AND, Thomas EJ. The frequency of diagnostic errors in outpatient care. *BMJ Qual Saf* 2014;23(9):727–731. doi:10.1136/bmjqs-2013-002627.

6. Slawomirski L, Kelly D, de Bienassis K, Kallas K-A, Klazinga N. The economics of diagnostic safety. *OECD Health Working Papers* 2025. doi:10.1787/fc61057a-en.

7. Rowley J. The wisdom hierarchy: representations of the DIKW hierarchy. *J Inf Sci* 2007;33(2):163–180. doi:10.1177/0165551506070706.

8. Li R, Niu Y, Scott SR, et al. Using Electronic Medical Record Data for Research in a HIMSS Analytics EMRAM Stage 7 Hospital in Beijing: Cross-sectional Study. *JMIR Med Inform* 2021;9(8):e24405. doi:10.2196/24405.

9. Duncan R, Eden R, Woods L, Wong I, Sullivan C. Synthesizing Dimensions of Digital Maturity in Hospitals: Systematic Review. *J Med Internet Res* 2022;24(3):e32994. doi:10.2196/32994.

10. Blumenthal D, Tavenner M. The "Meaningful Use" Regulation for Electronic Health Records. *N Engl J Med* 2010;363:501–504. doi:10.1056/NEJMp1006114.

11. Adler-Milstein J, et al. Electronic Health Record Adoption In US Hospitals: Progress Continues, But Challenges Persist. *Health Aff* 2015;34(12):2174–2180. doi:10.1377/hlthaff.2015.0992.

12. Weed LL. Medical Records That Guide and Teach. *N Engl J Med* 1968;278:593–600. doi:10.1056/NEJM196803212781204.

13. Bodenreider O, Cornet R, Vreeman DJ. Recent Developments in Clinical Terminologies: SNOMED CT, LOINC, and RxNorm. *Yearb Med Inform* 2018;27(1):129–139. doi:10.1055/s-0038-1667077.

14. Bender D, Sartipi K. HL7 FHIR: An Agile and RESTful Approach to Healthcare Information Exchange. *IEEE CBMS* 2013:326–331. doi:10.1109/CBMS.2013.6627810.

15. Mandel JC, Kreda DA, Mandl KD, Kohane IS, Ramoni RB. SMART on FHIR: a standards-based, interoperable apps platform for electronic health records. *J Am Med Inform Assoc* 2016;23(5):899–908. doi:10.1093/jamia/ocv189.

16. Friedman C, Alderson PO, Austin JHM, Cimino JJ, Johnson SB. A general natural-language text processor for clinical radiology (MedLEE). *J Am Med Inform Assoc* 1994;1(2):161–174. doi:10.1136/jamia.1994.95236146.

17. Savova GK, et al. Mayo clinical Text Analysis and Knowledge Extraction System (cTAKES). *J Am Med Inform Assoc* 2010;17(5):507–513. doi:10.1136/jamia.2009.001560.

18. Safran C, Bloomrosen M, Hammond WE, et al. Toward a National Framework for the Secondary Use of Health Data: An AMIA White Paper. *J Am Med Inform Assoc* 2007;14(1):1–9. doi:10.1197/jamia.M2273.

19. Rosenbloom ST, Denny JC, Xu H, et al. Data from clinical notes: a perspective on the tension between structure and flexible documentation. *J Am Med Inform Assoc* 2011;18(2):181–186. doi:10.1136/jamia.2010.007237.

20. Peleg M. Computer-interpretable clinical guidelines: a methodological review. *J Biomed Inform* 2013;46(4):744–763. doi:10.1016/j.jbi.2013.06.009.

21. Lenz R, Reichert M. IT support for healthcare processes – premises, challenges, perspectives. *Data Knowl Eng* 2007;61(1):39–58. doi:10.1016/j.datak.2006.04.007.

22. van der Aalst W. *Process Mining: Data Science in Action*. 2nd ed. Berlin: Springer; 2016. doi:10.1007/978-3-662-49851-4.

23. Rojas E, Munoz-Gama J, Sepúlveda M, Capurro D. Process mining in healthcare: A literature review. *J Biomed Inform* 2016;61:224–236. doi:10.1016/j.jbi.2016.04.007.

24. Kanaparthy NS, Bakare T, Diao Z, et al. Real-World Evidence Synthesis of Digital Scribes Using Ambient Listening and Generative Artificial Intelligence for Clinician Documentation Workflows: Rapid Review. *JMIR AI* 2025;4:e76743. doi:10.2196/76743.

25. Soysal E, et al. CLAMP: a toolkit for efficiently building customized clinical NLP pipelines. *J Am Med Inform Assoc* 2018;25(3):331–336. doi:10.1093/jamia/ocx132.

26. Ji Z, et al. Survey of Hallucination in Natural Language Generation. *ACM Comput Surv* 2023;55(12):248. doi:10.1145/3571730.

27. Nori H, et al. Capabilities of GPT-4 on Medical Challenge Problems. *arXiv* 2023:2303.13375.

28. Garcez A, Lamb LC. Neurosymbolic AI: the 3rd wave. *Artif Intell Rev* 2023;56:12387–12406. doi:10.1007/s10462-023-10448-w.

29. Uzuner O, South BR, Shen S, DuVall SL. 2010 i2b2/VA challenge on concepts, assertions, and relations in clinical text. *J Am Med Inform Assoc* 2011;18(5):552–556. doi:10.1136/amiajnl-2011-000203.

30. Sun W, Rumshisky A, Uzuner O. Evaluating temporal relations in clinical text: 2012 i2b2 Challenge. *J Am Med Inform Assoc* 2013;20(5):806–813. doi:10.1136/amiajnl-2013-001628.

31. Strötgen J, Gertz M. Multilingual and cross-domain temporal tagging (HeidelTime). *Lang Resour Eval* 2013;47(2):269–298. doi:10.1007/s10579-012-9179-y.

32. Chang AX, Manning CD. SUTime: A library for recognizing and normalizing time expressions. *Proc. LREC* 2012:3735–3740.

33. Goddard K, Roudsari A, Wyatt JC. Automation bias: a systematic review of frequency, effect mediators, and mitigators. *J Am Med Inform Assoc* 2012;19(1):121–127. doi:10.1136/amiajnl-2011-000089.

34. Aperstein Y, Gottlib A, Benita G, Apartsin A. Explainable Semantic Text Relations: A Question-Answering Framework for Comparing Document Content. *Information* 2025;16(12):1090. doi:10.3390/info16121090.

35. Laufer M, Aperstein Y, Apartsin A. Reliable Extraction of Clinical Follow-Up Instructions: A Hybrid Neural-Symbolic Pipeline. *arXiv*; 2026. arXiv:2605.26560.

36. Van Tiem J, Cramer E, Iverson C, et al. Listening to the note: clinician perspectives on ambient artificial intelligence scribes in medical documentation. *J Am Med Inform Assoc* 2026;33(2):255–262. doi:10.1093/jamia/ocaf214.